



Model Deployment Fundamentals

LECTURE

Deployment Methods



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In this lecture Deployment Methods, we discuss Gen AI model deployment and examine various deployment paradigms.

Gen AI Model Deployment

Deploying a GenAI/LLM model in a few words

The process of integrating a AI model into a **production environment**, making it accessible for end-users or other systems to **generate predictions** or **completions**.

(Deployment Strategies: [batch](#), [streaming](#), [real-time](#), or [embedded/edge](#))

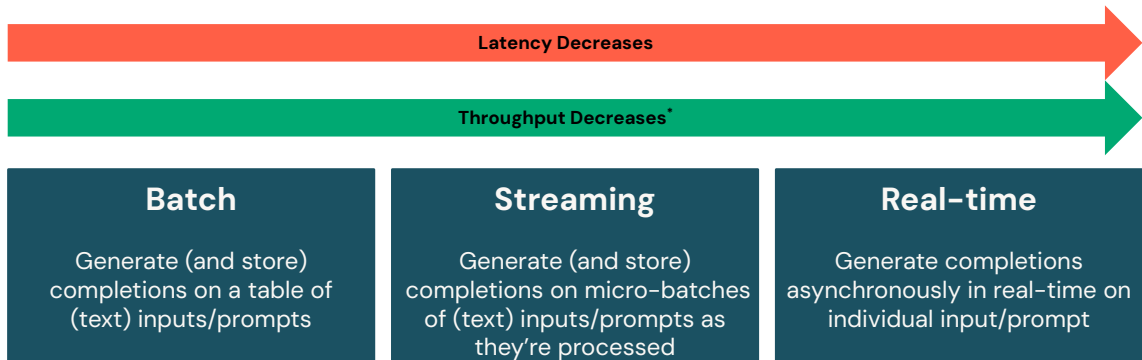


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Machine learning model deployment means integrating an AI model into a production environment so it can be used by end users or critical business applications. This allows the model to provide predictions, or for generative AI, produce text completions—usually in the form of textual outputs that support real-world systems and tasks. Deploying a machine learning model means putting the AI model into a production environment so it's accessible for end users or business systems. This way, applications can use the model's predictions or, for generative AI, its text completions to support business needs and workflows.

Deployment Paradigms

Similar to traditional machine learning extended to GenAI



Note: Edge (on-device) deployment is challenging with large language models due to space requirements.



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There are several ways to deploy machine learning models, based on how fast predictions are needed and how much data must be processed:

Batch deployment computes predictions ahead of time for large amounts of data. It runs at scheduled intervals (like hourly or daily), so latency is low and predictions aren't needed instantly. An example is summarizing long documents.

Streaming deployment processes data in micro-batches, with latency of seconds to minutes and moderate throughput. This fits tasks like translating product reviews as they come in.

Real-time deployment demands instant responses and high concurrency, typical for chatbots or RAG (Retrieval Augmented Generation) applications.

Edge deployment runs small models locally on devices, like cars, for simple tasks (not covered in detail here).

Each approach suits different business and technical needs, balancing speed, scale, and hardware requirements.

Deployment Paradigms

Comparing methods

Deployment Method	Throughput	Latency	Example Application
 Batch	High	High (<i>hours to days</i>)	Summarizing financial reports and generating insights.
Streaming	Moderate	Moderate (<i>seconds to minutes</i>)	Personalizing marketing messages.
 Real-time	Low-High	Low (<i>milliseconds</i>)	Chatbots (e.g. customer service, doc assistant)
Edge/Embedded	Low	Low (Dependent on device processing power)	Modify air conditioning temperature in a car using voice command



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To recap, model deployment methods include batch, streaming, real-time, and edge device deployments. For today's focus, you'll mainly work with batch and real-time approaches. To recap, the main deployment methods are batch, streaming, real-time, and edge device deployments. The focus here will mainly be on batch and real-time deployments. To recap, the different model deployment methods are batch, streaming, real-time, and edge device deployments. The main focus here will be on batch and real-time approaches. To recap, model deployment can be done via batch, streaming, real-time, or edge methods.



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